

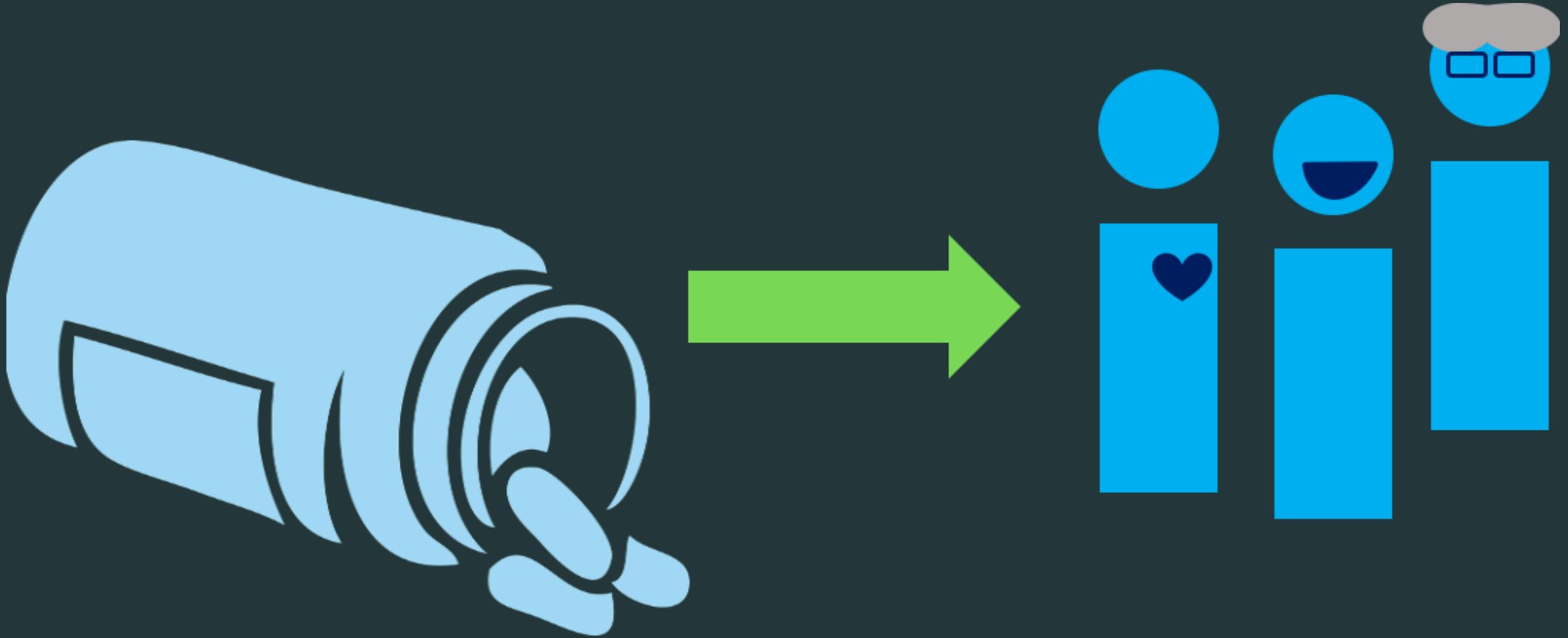
Causal Inference with group_by and summarise

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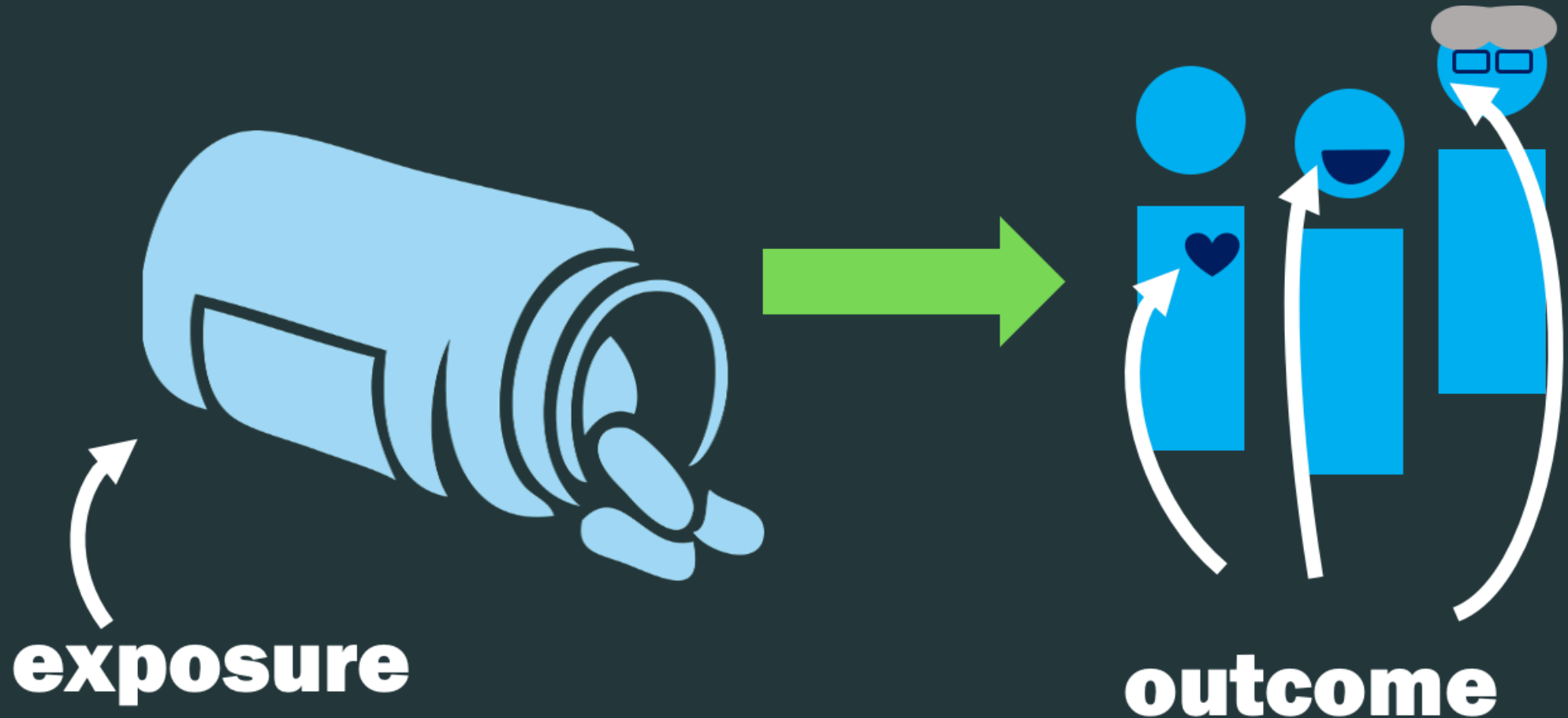
Observational Studies

Goal: To answer a research question



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Observational Studies

Randomized Controlled Trial

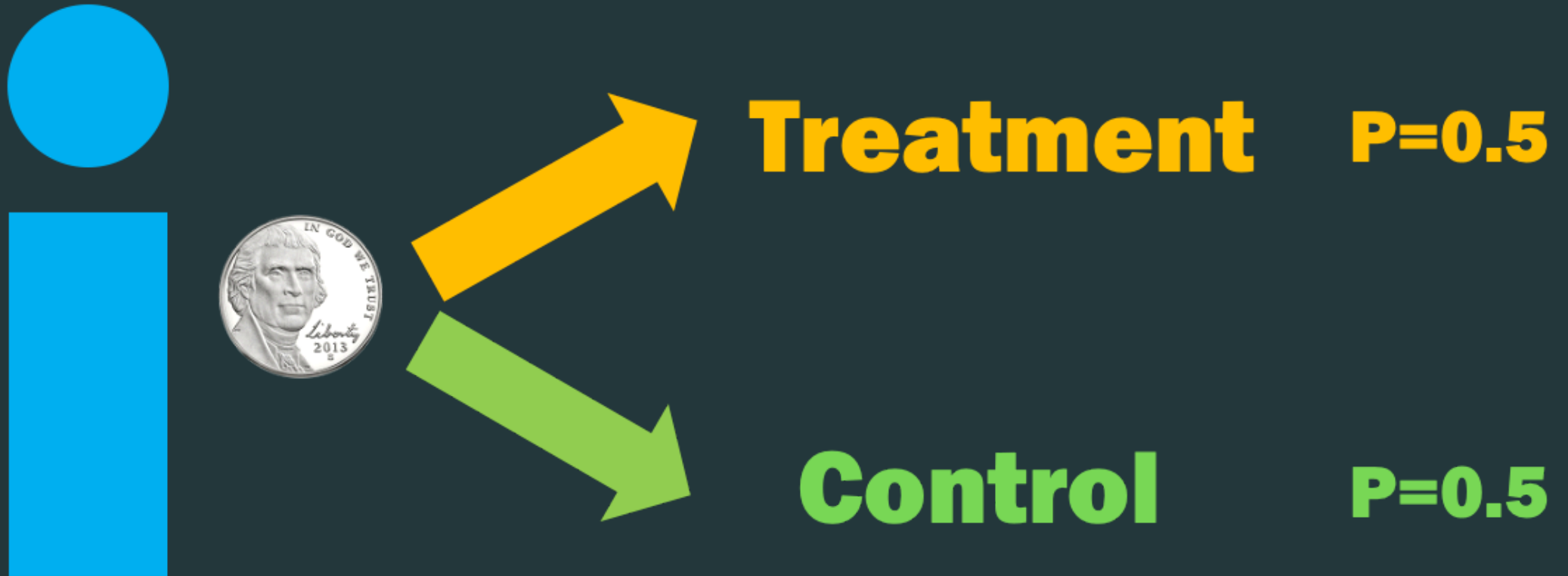


Treatment

Control

Observational Studies

Randomized Controlled Trial



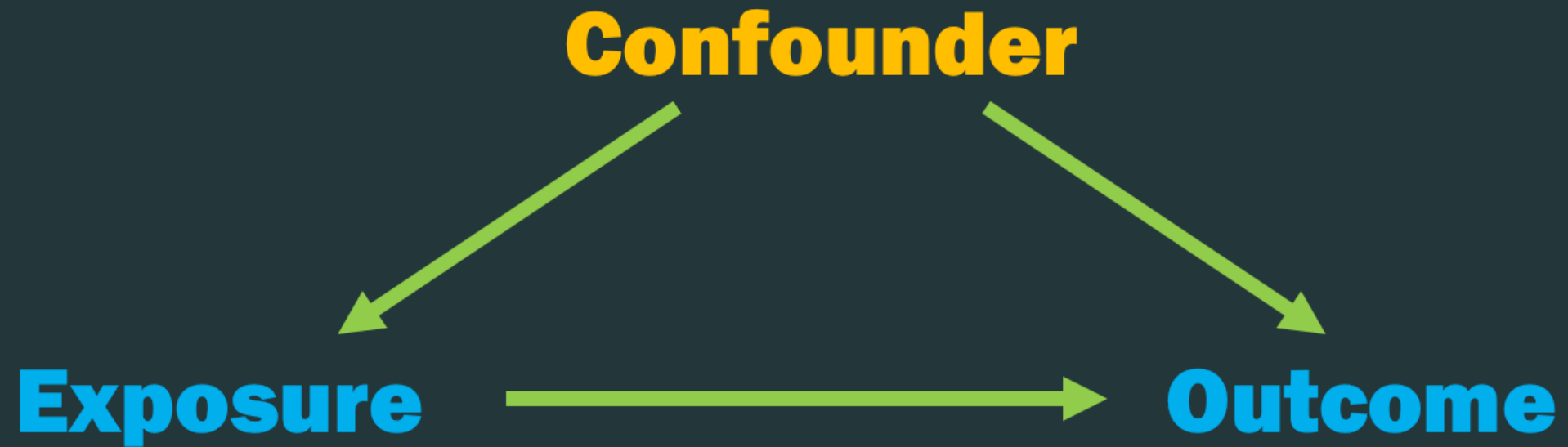
Observational Studies



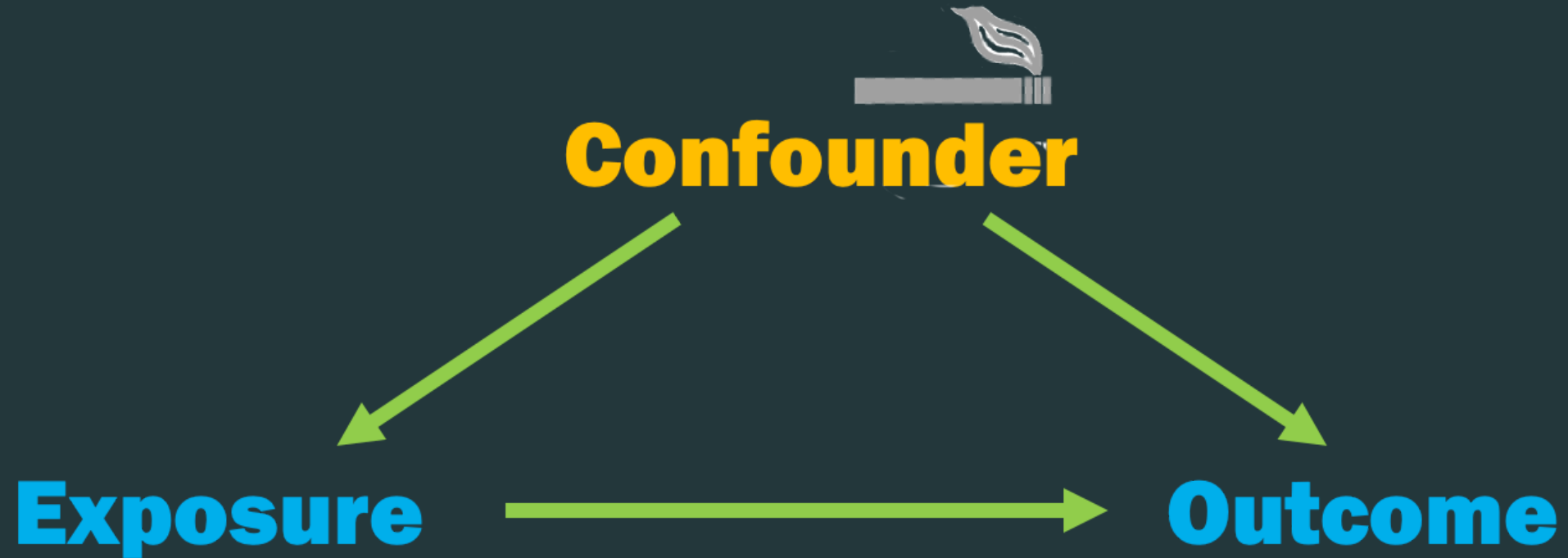




Confounding



Confounding



One binary confounder

Simulation

```
1 n <- 1000
2 sim <- tibble(
3   confounder = rbinom(n, 1, 0.5),
4   p_exposure = case_when(
5     confounder == 1 ~ 0.75,
6     confounder == 0 ~ 0.25
7   ),
8   exposure = rbinom(n, 1, p_exposure),
9   # potential outcomes: exposure is not in the
10  # outcome equation, so the true effect is 0
11  y0 = confounder + rnorm(n),
12  y1 = y0,
13  # we only ever observe one of the two
14  outcome = (1 - exposure) * y0 + exposure * y1
15 ) |> select(confounder, exposure, outcome)
16
17 sim
```

Simulation

```
# A tibble: 1,000 × 3
  confounder exposure outcome
  <int>         <int>     <dbl>
1         0         0     1.13
2         0         0     1.11
3         1         1     0.129
4         1         0     1.21
5         0         0     0.0694
6         1         1    -0.663
7         1         1     1.81
8         1         1    -0.912
9         1         0    -0.247
10        0         0     0.998
" : 999 more rows
```

Simulation

```
1 lm(outcome ~ exposure, data = sim)
```

Call:

```
lm(formula = outcome ~ exposure, data = sim)
```

Coefficients:

(Intercept)	exposure
0.2688	0.4070

- The true difference is 0, because exposure does not appear in the outcome equation

Simulation

```
1 sim |>
2   group_by(exposure) |>
3   summarise(avg_y = mean(outcome))
```

```
# A tibble: 2 × 2
  exposure avg_y
  <int> <dbl>
1       0 0.269
2       1 0.676
```

Simulation

```
1 sim |>
2   group_by(exposure) |>
3   summarise(avg_y = mean(outcome)) |>
4   pivot_wider(
5     names_from = exposure,
6     values_from = avg_y,
7     names_prefix = "x_"
8   ) |>
9   summarise(estimate = x_1 - x_0)
```

```
# A tibble: 1 × 1
  estimate
  <dbl>
1    0.407
```

Your Turn 1 (04-ci-with-group-by-and-summarise-exercises.qmd)

Group the dataset by `confounder` and `exposure`

Calculate the mean of the `outcome` for the groups

Your Turn 1

```
1 sim |>
2   group_by(confounder, exposure) |>
3   summarise(avg_y = mean(outcome), .groups = "drop")
```

A tibble: 4 × 3

	confounder <int>	exposure <int>	avg_y <dbl>
1	0	0	-0.00907
2	0	1	-0.0166
3	1	0	1.09
4	1	1	0.936

Your Turn 1

```
1 sim |>
2   group_by(confounder, exposure) |>
3   summarise(avg_y = mean(outcome), .groups = "drop") |>
4   pivot_wider(
5     names_from = exposure,
6     values_from = avg_y,
7     names_prefix = "x_"
8   ) |>
9   reframe(estimate = x_1 - x_0) |>
10  # note: we would need to weight this
11  # if the confounder groups were not equal sized
12  summarise(estimate = mean(estimate))
```

```
# A tibble: 1 × 1
  estimate
  <dbl>
1 -0.0794
```



Two binary confounders

Simulation

```
1 n <- 1000
2 sim2 <- tibble(
3   confounder_1 = rbinom(n, 1, 0.5),
4   confounder_2 = rbinom(n, 1, 0.5),
5   p_exposure = case_when(
6     confounder_1 == 1 & confounder_2 == 1 ~ 0.75,
7     confounder_1 == 0 & confounder_2 == 1 ~ 0.9,
8     confounder_1 == 1 & confounder_2 == 0 ~ 0.2,
9     confounder_1 == 0 & confounder_2 == 0 ~ 0.1,
10  ),
11  exposure = rbinom(n, 1, p_exposure),
12  outcome = confounder_1 + confounder_2 + rnorm(n)
13 ) |> select(-p_exposure)
14
15 sim2
```

Simulation

```
# A tibble: 1,000 × 4
```

	confounder_1 <int>	confounder_2 <int>	exposure <int>	outcome <dbl>
1	0	0	0	0.521
2	1	0	0	1.38
3	0	0	0	-0.624
4	0	1	1	0.427
5	1	0	1	1.31
6	0	0	0	-0.707
7	1	1	1	2.52
8	1	0	0	1.45
9	0	0	0	-0.505
10	0	1	1	0.793

```
"# A tibble: 1,000 × 4"
```

Simulation

```
1 lm(outcome ~ exposure, data = sim2)
```

Call:

```
lm(formula = outcome ~ exposure, data = sim2)
```

Coefficients:

(Intercept)	exposure
0.6395	0.6951

- The true difference is again 0, because exposure does not appear in the outcome equation

Your Turn 2

Group the dataset by the confounders and exposure

Calculate the mean of the outcome for the groups

Your Turn 2

```
1 sim2 |>
2   group_by(confounder_1, confounder_2, exposure) |>
3   summarise(avg_y = mean(outcome), .groups = "drop") |>
4   pivot_wider(
5     names_from = exposure,
6     values_from = avg_y,
7     names_prefix = "x_"
8   ) |>
9   reframe(estimate = x_1 - x_0) |>
10  summarise(estimate = mean(estimate))
```

```
# A tibble: 1 × 1
  estimate
  <dbl>
1 -0.0731
```

Simulation

```
1 n <- 100000
2 big_sim2 <- tibble(
3   confounder_1 = rbinom(n, 1, 0.5),
4   confounder_2 = rbinom(n, 1, 0.5),
5   p_exposure = case_when(
6     confounder_1 == 1 & confounder_2 == 1 ~ 0.75,
7     confounder_1 == 0 & confounder_2 == 1 ~ 0.9,
8     confounder_1 == 1 & confounder_2 == 0 ~ 0.2,
9     confounder_1 == 0 & confounder_2 == 0 ~ 0.1,
10  ),
11   exposure = rbinom(n, 1, p_exposure),
12   outcome = confounder_1 + confounder_2 + rnorm(n)
13 ) |> select(-p_exposure)
14
15 big_sim2
```

Simulation

```
# A tibble: 100,000 × 4
```

	confounder_1 <int>	confounder_2 <int>	exposure <int>	outcome <dbl>
1	1	1	1	2.35
2	1	1	0	3.71
3	0	0	0	2.08
4	0	1	1	0.516
5	0	0	0	-0.166
6	1	1	1	1.58
7	0	0	0	0.472
8	1	0	0	3.22
9	0	1	1	0.929
10	0	1	1	1.41

```
"# A tibble: 100,000 × 4"
```

Simulation

```
1 lm(outcome ~ exposure, data = big_sim2)
```

Call:

```
lm(formula = outcome ~ exposure, data = big_sim2)
```

Coefficients:

(Intercept)	exposure
0.6782	0.6561

Simulation

```
1 big_sim2 |>
2   group_by(confounder_1, confounder_2, exposure) |>
3   summarise(avg_y = mean(outcome), .groups = "drop") |>
4   pivot_wider(
5     names_from = exposure,
6     values_from = avg_y,
7     names_prefix = "x_"
8   ) |>
9   reframe(estimate = x_1 - x_0) |>
10  summarise(estimate = mean(estimate))
```

```
# A tibble: 1 × 1
  estimate
  <dbl>
1  0.0187
```

Continuous confounder?

Simulation

```
1 n <- 10000
2 sim3 <- tibble(
3   confounder = rnorm(n),
4   p_exposure = exp(confounder) / (1 + exp(confounder)),
5   exposure = rbinom(n, 1, p_exposure),
6   outcome = confounder + rnorm(n)
7 ) |> select(-p_exposure)
8
9 sim3
```

Simulation

```
# A tibble: 10,000 × 3
  confounder exposure outcome
    <dbl>      <int>    <dbl>
1   -0.167         0   -0.560
2    0.252         1    0.628
3   -0.321         1   -0.608
4    0.621         0    1.58
5   -0.619         1    0.358
6   -0.897         0   -1.95
7   -2.01         0   -2.50
8    0.296         0   -1.10
9   -0.504         1   -0.316
10  -0.536         1    1.12
```

Simulation

```
1 lm(outcome ~ exposure, data = sim3)
```

Call:

```
lm(formula = outcome ~ exposure, data = sim3)
```

Coefficients:

(Intercept)	exposure
-0.4036	0.8152

- The true difference is still 0, because exposure does not appear in the outcome equation

Your Turn 3

Use `ntile()` from `dplyr` to calculate a binned version of `confounder` called `confounder_q`. We'll create a variable with 5 bins.

Group the dataset by the binned variable you just created and exposure

Calculate the mean of the outcome for the groups

Your Turn 3

```
1 sim3 |>
2   mutate(confounder_q = ntile(confounder, 5)) |>
3   group_by(confounder_q, exposure) |>
4   summarise(avg_y = mean(outcome), .groups = "drop") |>
5   pivot_wider(
6     names_from = exposure,
7     values_from = avg_y,
8     names_prefix = "x_"
9   ) |>
10  reframe(estimate = x_1 - x_0) |>
11  summarise(estimate = mean(estimate))
```

```
# A tibble: 1 × 1
  estimate
  <dbl>
1  0.0728
```

How many bins?

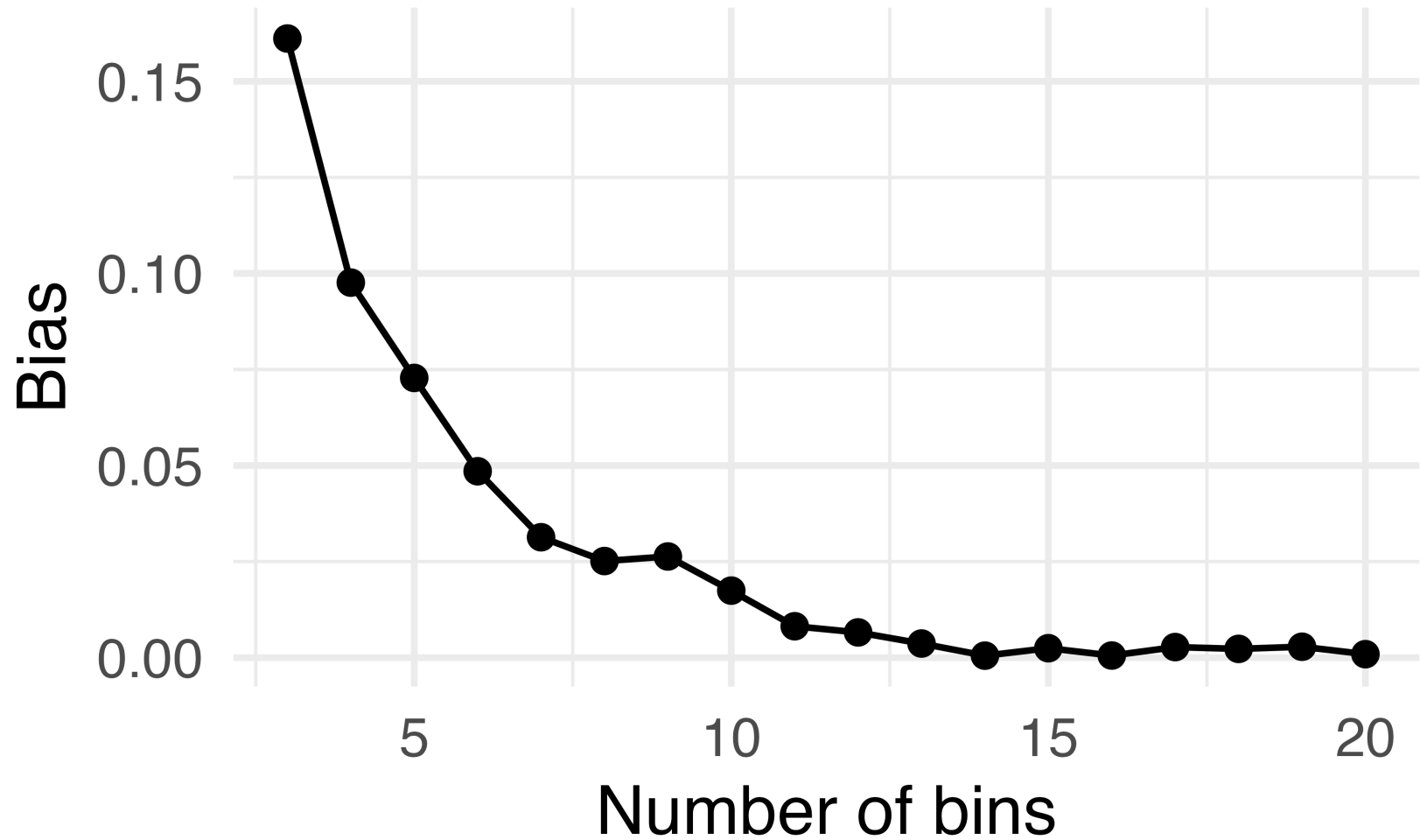
```
1 bin_estimate <- function(bins) {  
2   sim3 |>  
3   mutate(confounder_q = ntile(confounder, bins)) |>  
4   group_by(confounder_q, exposure) |>  
5   summarise(avg_y = mean(outcome), .groups = "drop") |>  
6   pivot_wider(  
7     names_from = exposure,  
8     values_from = avg_y,  
9     names_prefix = "x_"  
10  ) |>  
11  reframe(estimate = x_1 - x_0) |>  
12  summarise(bins = bins, estimate = mean(estimate))  
13 }
```

How many bins?

```
1 map(3:20, bin_estimate) |>
2   bind_rows() |>
3   ggplot(aes(x = bins, y = abs(estimate))) +
4   geom_point() +
5   geom_line() +
6   labs(y = "Bias", x = "Number of bins") +
7   theme_minimal(base_size = 20)
```

- As we increase the number of bins, the bias decreases

How many bins?



How many bins?

```
1 bin_estimate(500)
```

```
# A tibble: 1 × 2  
  bins estimate  
  <dbl>     <dbl>  
1    500         NA
```

- The estimate is NA

A positivity violation

- Some bins have no one in one of the exposure groups, making their difference inestimable
- The violation is stochastic: it comes from our sample size, 10,000, and the number of bins, 500

A parametric alternative

- This non-parametric method, while flexible, is limited by our sample size
- Parametric models let us extrapolate under certain assumptions

A parametric alternative

```
1 lm(outcome ~ exposure + confounder, data = sim3)
```

Call:

```
lm(formula = outcome ~ exposure + confounder, data = sim3)
```

Coefficients:

(Intercept)	exposure	confounder
0.01853	-0.01884	1.01294

- We get the right answer without binning, at the price of the correct **functional form**

Stratification

- Grouping by confounders is called **stratification**, a **non-parametric** approach
- We aren't using a statistical model to restrict the form of the variables

The curse of dimensionality

- With many confounders, especially continuous ones, we have too few observations within combinations of confounder levels
- We saw a version of this with 500 bins: no data means no estimate

What if we could come
up with a **summary**
score of all
confounders?